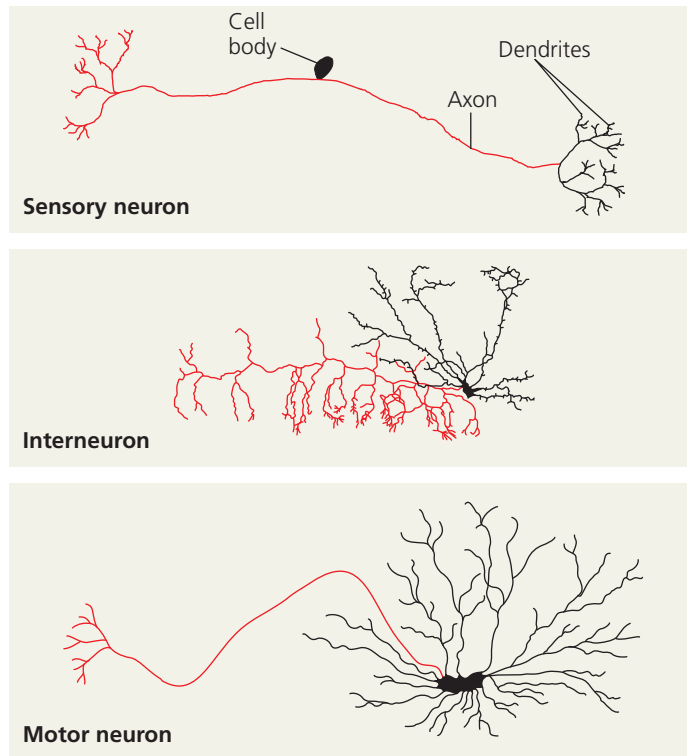


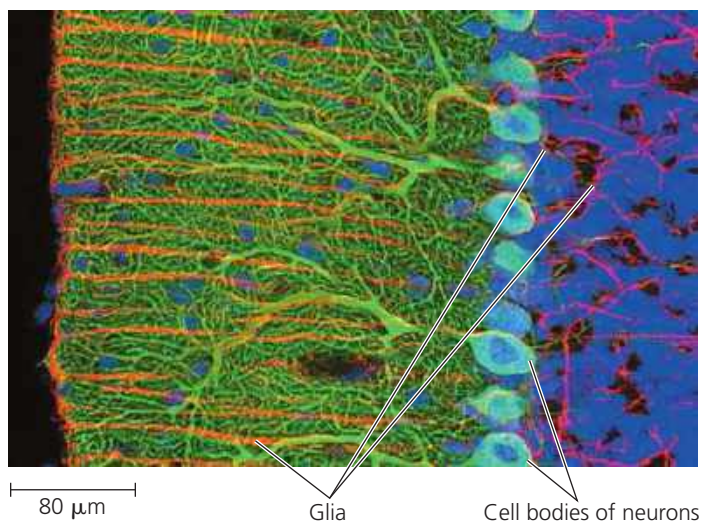
information along its length in the same way as a neuron that controls the movement of a body part. The particular connections made by the active neuron are what distinguish the type of information being transmitted. Interpreting nerve impulses therefore involves sorting neuronal paths and connections.

As shown in **Figure 48.4**, the shape of a neuron can vary from simple to quite complex, depending on its role in

▼ **Figure 48.4 Structural diversity of neurons.** In these drawings of neurons, cell bodies and dendrites are black and axons are red.



▼ **Figure 48.5 Glia in the mammalian brain.** This fluorescently labeled laser confocal micrograph shows a region of the rat brain packed with glia and interneurons. The glia are labeled red, the DNA in nuclei is labeled blue, and the dendrites of neurons are labeled green.



information processing. Neurons that have highly branched dendrites, such as some interneurons, can receive input through tens of thousands of synapses. Similarly, neurons that transmit information to many target cells do so through highly branched axons. When grouped together, the axons of neurons form the bundles we call **nerves**.

In many animals, the neurons that carry out sorting, processing, and integration are organized in a **central nervous system (CNS)**, which may include a **brain** or simpler clusters called **ganglia**. The neurons that carry information into and out of the CNS constitute the **peripheral nervous system (PNS)**. Neurons of both the CNS and PNS require supporting cells called glial cells, or **glia** (from a Greek word meaning “glue”) (**Figure 48.5**).

CONCEPT CHECK 48.1

1. Compare and contrast the structure and function of axons and dendrites.
2. Describe the basic pathway of information flow through neurons that causes you to turn your head when someone calls your name.
3. **WHAT IF?** How might increased branching of an axon help coordinate responses to signals communicated by the nervous system?

For suggested answers, see Appendix A.

CONCEPT 48.2

Ion pumps and ion channels establish the resting potential of a neuron

We turn now to the essential role of ions in neuronal signaling. In neurons, as in other cells, ions are unequally distributed between the interior of cells and the surrounding fluid (see Concept 7.4). As a result, the inside of a cell is negatively charged relative to the outside. This charge difference, or **voltage**, across the plasma membrane is called the **membrane potential**, reflecting the fact that the attraction of opposite charges across the plasma membrane is a source of potential energy. For a resting neuron—one that is not sending a signal—the membrane potential is called the **resting potential** and is typically between -60 and -80 millivolts (mV).

When a neuron receives a stimulus, the membrane potential changes. Rapid shifts in membrane potential are what enable us to see the intricate structure of a spiderweb, hear a song, or ride a bicycle. These changes, which are known as **action potentials**, will be discussed in Concept 48.3. To understand how they convey information, we need to explore the ways in which membrane potentials are formed, maintained, and altered.